Lecture 1: Course introduction

PSYC 51.07: Models of language and communication

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About the instructor

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ContextLab

Research focus

How do our brains support our ongoing conscious thoughts, and how (and what) do we remember?

Key areas

Learning and memory, education technology, brain network dynamics, data science, NLP

Approach

Theory, models, experiments, neuroimaging

Training

- B.S., Neuroscience & Computer Science
- Ph.D., Neuroscience
- Postdoc, Computer Science & Neuroscience

Funding & collaborators

NIH National Institutes of Health

NSF

DARPA

intel labs

Meta

Google

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What is this course about?

Course content

We will explore how machines can understand and generate human language:

- Building conversational agents from scratch
- Understanding how language relates to thought
- Hands-on programming with real models
- Critical thinking about AI consciousness

Course design

This course is experiential! You will learn by doing: coding, experimenting, discussing, and researching.

Course structure

ELIZA → SPAM → Wikipedia → Chatbot → GPT

We'll build from pattern matching to transformer-based models through our 5 main projects

Grading

- Bi-weekly short projects (5): 75%
- Final project: 25%
- Can work individually or in groups
- See syllabus for additional details

Tools

- Google Colaboratory
- HuggingFace
- GitHub / Discord
- GenAI (Claude, ChatGPT, Gemini)

The big questions

1. Can machines truly *understand* language?
2. What is the relationship between **language** and **thought**?
3. Can statistical patterns capture **meaning**? How? Under which circumstances?
4. Can AI be conscious? If so, what are the implications?

> **Note**
>
> These are not just "fluff" questions: they are at the heart of cognitive science, philosophy, psychology, and neuroscience!

Discussion: Is ChatGPT conscious?

- What does "conscious" even mean?
- How would we test for consciousness?
- Does it matter if ChatGPT *seems* conscious?

For further consideration

What are the implications for

- Ourselves
- Other animals
- Other life forms in general (aliens? synthetic life?)
- Policy, ethics, and society more broadly

What is consciousness?

- **Phenomenal:** Subjective experience
- **Access:** Information available for reasoning
- **Self-awareness:** Knowledge of one's own mental states

Example of phenomenal consciousness

The redness of red, pain, taste of coffee

Example of access consciousness

Being able to report on and use information to guide behavior

Example of self-awareness

Knowing that you are thinking, or recognizing your own emotions

For further consideration

If ChatGPT says "I feel happy," does it actually *feel* anything?

The hard problem

With humans, we assume consciousness because:

- We share similar biology
- We have our own conscious experience
- They behave consistently with having experiences

But with AI:

- Completely different "biology" (silicon vs. neurons)
- No shared evolutionary history
- Can produce human-like behavior

Note

Why is this so difficult? We cannot directly observe consciousness--even in other humans!

The Chinese room argument

Note

John Searle (1980): Thought experiment about understanding vs. simulation

The setup: Person in room with Chinese symbols and rule book

Chinese question → Rule book → Chinese answer

Person follows rules but does not understand Chinese

Tip

Does following rules = understanding? Searle argues: **No!**

Current scientific consensus

> Warning
>
> **Important:** Most cognitive scientists and AI researchers agree: **Current LLMs are not conscious.**

Why not?

- No sensory-motor grounding in the world
- No persistent self-model or goals
- Pattern matching $\neq$ understanding

Language and thought

> **Note**
>
> **Discussion:** Do you need language to think? Does language *shape* how you think?

Language = Thought:
All thinking happens in language

Language $\neq$ Thought:
Language is just a tool for communication

The truth is likely somewhere in between…

The language-thought spectrum

Strong Whorfian → Weak Whorfian → Moderate → Language-of-Thought

Language shapes thought (left) to language independent of thought (right)

Current evidence points toward the middle: language and thought interact in complex ways, but are not identical.

Evidence: The language network

Note

Fedorenko et al. (2024) - Nature: *The language network as a natural kind*

Key findings:

- The brain has a **specialized language network**
- Distinct from: reasoning, math, social cognition, music

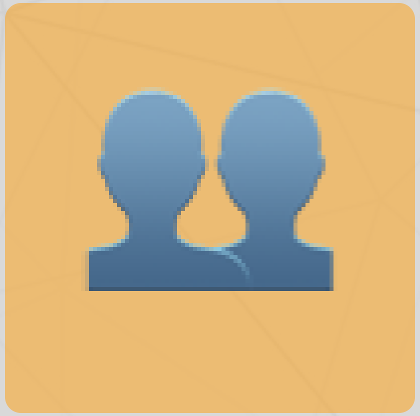
Tip

Implication: Language and thought are *separable* in the brain!

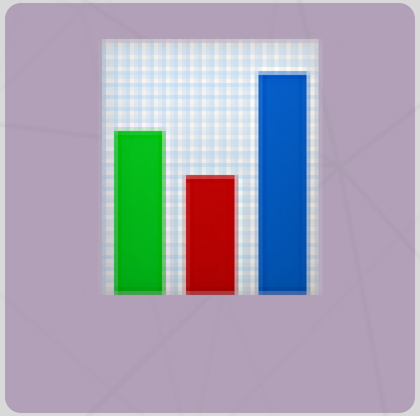
Brain networks

Language

Reasoning



Social



Math

> **Note**
>
> **Key insight:** Language is a specialized system, not the basis of all thought!

Evidence: Language shapes perception

Note

Lupyan et al. (2020) - TICS: *Effects of language on visual perception*

Key findings: Having words for things affects how we *see* them

- Speed up visual search
- Alter color perception
- Influence object categorization

The Russian blue example

English speakers:

- One word: "blue"
- Slower to distinguish shades

Russian speakers:

- "siniy" (dark) vs "goluboy" (light)
- Faster to distinguish shades

Tip

Language categories affect *perception*, not just description!

So what about LLMs?

Current evidence suggests:

- LLMs are *very good* at language patterns
- LLMs may have some internal "representations"
- LLMs lack grounding in sensory-motor experience
- No evidence of phenomenal consciousness

Tip

Can an LLM have sophisticated language *without* sophisticated thought?

The grounding problem

> **Note**
>
> **Symbol grounding:** How do symbols (words) get their meaning?

Humans: Grounded in experience

- See, touch, taste objects
- Act in the world

LLMs: Statistical patterns

- Only see text
- No sensory experience

Our approach

Note

Philosophy of this course: We will build language models *from scratch* to understand what they can and cannot do.

Critical perspective:

- Question assumptions about "understanding"
- Appreciate both capabilities and limitations
- Think about implications for cognitive science

From simple to complex

ELIZA → Classifiers → Embeddings → Attention → GPT

At what point does pattern matching become understanding?

> **Note**
>
> **Discussion:** At what point (if any) does pattern matching become understanding?

HuggingFace: Your learning companion

We will use HuggingFace for:

- **Course materials:** Free NLP course at huggingface.co/course
- **Pre-trained models:** Download and use state-of-the-art models
- **Datasets:** Access curated datasets for training

> Warning
>
> **Assignment:** Create a free HuggingFace account and complete Chapter 1 of the NLP course!

Next lectures this week

re 2: Pattern matching and ELIZA

ing manipulation

gular expressions

roduction to ELIZA

Note

Lecture 3 (X-hour): ELIZA implemen

- Complete architecture
- The ELIZA effect
- Assignment 1 details

Readings for this week

Note

Required readings:

1. Weizenbaum (1966): ELIZA
2. Fedorenko et al. (2024): The language network
3. Lupyan et al. (2020): Effects of language on visual perception

Tip: Start with Weizenbaum--it will help you understand the fundamentals!

Key takeaways

1. **Consciousness is complex:** Multiple types, hard to define
2. **Language $\neq$ Thought:** But they interact in interesting ways
3. **LLMs are not conscious:** They are sophisticated pattern matchers
4. **Grounding matters:** Meaning comes from experience
5. **We will build to understand:** Hands-on reveals true capabilities

Questions?

Email

Discord

Moore 349

Note

Office hours: By appointment | **Lab website:** context-lab.com